

作业

- 7.21 A CS amplifier utilizes a MOSFET biased at $I_D = 0.25$ mA with $V_{OV} = 0.25$ V and $R_D = 20$ k Ω . The amplifier is fed with a signal source having $R_{sig} = 100$ k Ω , and a 20-k Ω load is connected to the output. Find R_{in} , A_{vo} , R_o , A_v , and G_v . If, to maintain reasonable linearity, the peak of the input sine-wave signal is limited to 10% of $2V_{OV}$, what is the peak of the sine-wave voltage at the output?
Ans. ∞ ; -40 V/V; 20 k Ω ; -20 V/V; -20 V/V; 1 V

$$R_{in} = \infty$$

$$g_m = \frac{I_D}{V_{OV}/2} = 2 \text{ mA/V}$$

$$R_o = R_D = 20 \text{ k}\Omega$$

$$A_{vo} = -g_m R_D = -40 \text{ V/V}$$

$$A_v = -g_m (R_D || R_L) = -20 \text{ V/V}$$

$$G_v = A_v = -20 \text{ V/V}$$

$$\hat{v}_{in} = 0.1 \times 0.5 \text{ V} = 50 \text{ mV}$$

$$\hat{v}_{out} = \hat{v}_{in} \times 20 = 1 \text{ V}$$

7.24 Show that with R_e included, and v_π limited to a maximum value $\hat{v}_\pi$, the maximum allowable input signal, $\hat{v}_{sig}$, is given by

$$\hat{v}_{sig} = \hat{v}_\pi \left(1 + \frac{R_e}{r_e} + \frac{R_{sig}}{r_\pi} \right)$$

If the transistor is biased at $I_c = 0.5$ mA and has a β of 100, what value of R_e is needed to permit an input signal $\hat{v}_{sig}$ of 100 mV from a source with a resistance $R_{sig} = 10$ k Ω while limiting $\hat{v}_\pi$ to 10 mV? What is R_{in} for this amplifier? If the total resistance in the collector is 10 k Ω , what G_v value results?

Ans. 350 Ω ; 40.4 k Ω ; -19.8 V/V

$$g_m = \frac{I_c}{V_T} = \frac{0.5 \text{ mA}}{25 \text{ mV}} = 20 \text{ mA/V}$$

$$100 = 10 \left(1 + \frac{R_e}{\frac{1}{g_m}} + \frac{10 \text{ k}\Omega}{\frac{1}{g_m}} \right) \Rightarrow R_e = 350 \Omega$$

$$R_{in} = \frac{\beta}{g_m} + (\beta + 1) \cdot R_e = 40.35 \text{ k}\Omega$$

$$G_v = \frac{v_e \cdot (R_c \parallel R_L)}{v_b \cdot R_{sig} + v_b \cdot R_{in}} = \beta \cdot \frac{R_c \parallel R_L}{R_{sig} + R_{in}} = -19.86 \text{ V/V}$$

$$\text{or } G_v = \frac{R_{in}}{R_{sig} + R_{in}} \cdot \frac{-g_m R_c \parallel R_L}{1 + g_m R_e} = -20.03 \text{ V/V} \quad (\text{if } i_c \approx i_e \text{ is ok})$$

7.25 A CG amplifier is required to match a signal source with $R_{sig} = 100 \Omega$. At what current I_D should the MOSFET be biased if it is operated at an overdrive voltage of 0.20 V? If the total resistance in the drain circuit is 2 k Ω , what is the overall voltage gain?

Ans. 1 mA; 10 V/V

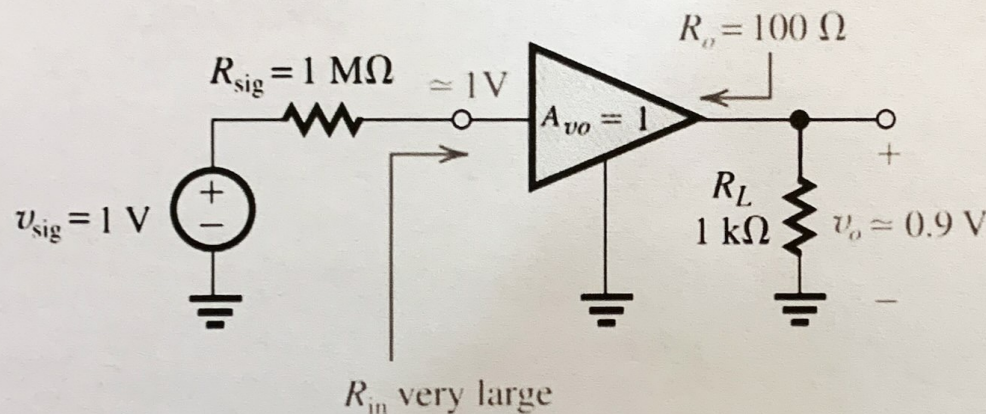
$$\text{match means } R_{sig} = R_{in} \Rightarrow 100 = \frac{1}{g_m} \Rightarrow g_m = \frac{1}{100} \text{ A/V}$$

$$g_m = \frac{I_D}{V_{ov}/2} \Rightarrow I_D = g_m \cdot \frac{V_{ov}}{2} = \frac{1}{100} \times 0.1 = 1 \text{ mA}$$

$$G_v = \frac{R_{in}}{R_{sig} + R_{in}} \cdot g_m (R_D \parallel R_L)$$

$$= \frac{1}{2} \cdot \frac{1}{100} \cdot 2\text{k} = 10 \text{ V/V}$$

- D7.28** It is required to design a source follower that implements the buffer amplifier shown in Fig. 7.42(c). If the MOSFET is operated with an overdrive voltage $V_{ov} = 0.25$ V, at what drain current should it be biased? Find the output signal amplitude and the signal amplitude between gate and source.
 Ans. 1.25 mA; 0.91 V; 91 mV



$$\frac{1}{g_m} = 100 \Rightarrow g_m = \frac{1}{100} \text{ A/V}$$

$$g_m = \frac{I_D}{V_{ov}/2} \Rightarrow I_D = \frac{1}{100} \cdot \frac{0.25}{2} = 1.25 \text{ mA}$$

$$v_{out} = \frac{R_L}{\frac{1}{g_m} + R_L} \cdot v_{in} = \frac{1}{0.1 + 1} = 0.91 \text{ V}$$

$$v_{gs} = v_{in} \cdot \frac{\frac{1}{g_m}}{\frac{1}{g_m} + R_L} = 91 \text{ mV}$$

7.30 An emitter follower uses a transistor with $\beta = 100$ and is biased at $I_C = 5 \text{ mA}$. It operates between a source with a resistance of $10 \text{ k}\Omega$ and a load of $1 \text{ k}\Omega$. Find R_{in} , G_{vo} , R_{out} , and G_v . What is the peak amplitude of v_{sig} that results in v_o having a peak amplitude of 5 mV ? Find the resulting peak amplitude at the output.

Ans. $101.5 \text{ k}\Omega$; 1 V/V ; $104 \text{ }\Omega$; 0.91 V/V ; 1.1 V ; 1 V

$$R_{in} = \left(\frac{\partial}{g_m} + r_e \right) \cdot (\beta + 1) \quad g_m = \frac{I_C}{V_T} = \frac{5 \text{ mA}}{25 \text{ mV}} = \frac{1}{5} \text{ A/V}$$

$$= \left(\frac{100}{101} \cdot 5 + 1 \text{ k} \right) \cdot 101$$

$$= 101.5 \text{ k}\Omega$$

$$G_{vo} = 1 \text{ V/V}$$

$$R_{out} = \frac{\partial}{g_m} + \frac{R_{sig}}{\beta + 1} = \frac{100}{101} \cdot 5 + \frac{10 \text{ k}}{101} = 104 \text{ }\Omega$$

$$G_v = \frac{v_o \cdot r_e}{v_b \cdot R_{sig} + v_b \cdot R_{in}} = 101 \cdot \frac{1 \text{ k}}{101.5 \text{ k} + 10 \text{ k}} = 0.91 \text{ V/V}$$

$$\hat{v}_{sig} = \underbrace{v_b + v_e \cdot \frac{\partial}{g_m}}_{\text{}} \left(r_e + \frac{R_{sig}}{\beta + 1} \right) = 5 \text{ mV} + \frac{5 \text{ mV} \cdot 101}{100 \cdot 5} \left(1 \text{ k} + \frac{10 \text{ k}}{101} \right) = 1.115 \text{ V}$$

$$\hat{v}_{out} = \hat{v}_{sig} \cdot G_v = 1.01 \text{ V}$$